



mendix

AI Experience



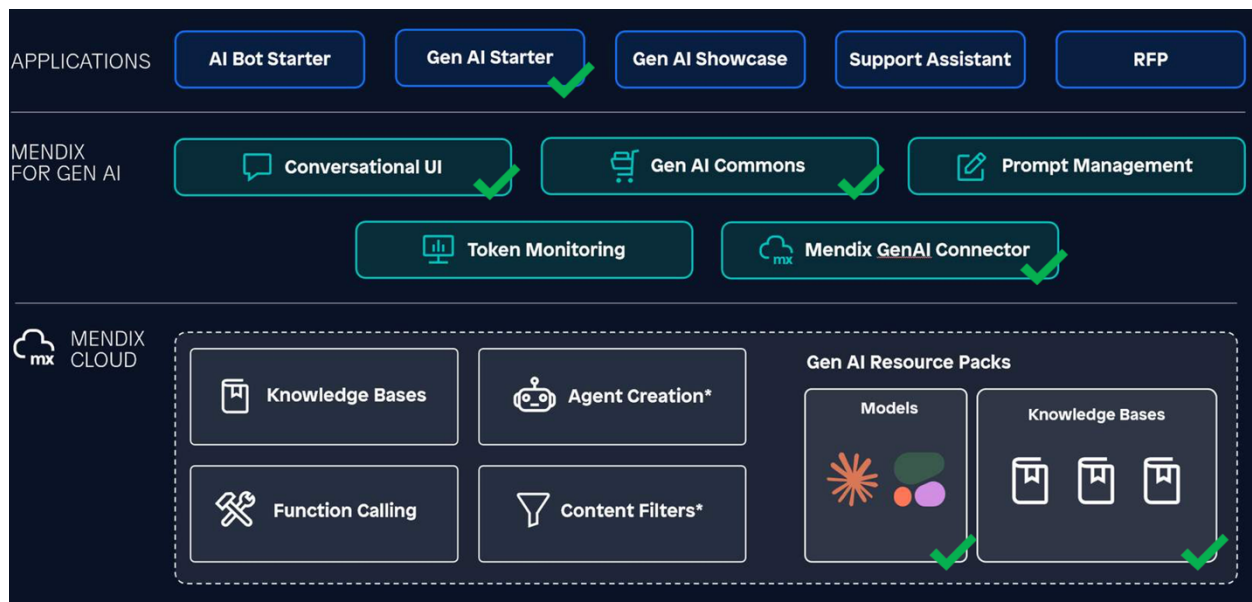
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1 Goal of this document

In this walkthrough you will extend existing application and get acquainted with:

- Mendix Cloud GenAI Resource Packs (Model, Embedding and Knowledge Base)
- Mendix Conversational UI
- Mendix GenAI Connector
- Retrieval Augmented Generation (RAG) with Mendix Cloud GenAI Resource Packs: Mendix Knowledge Base, Embedding and Text Generation



2 Main prerequisites

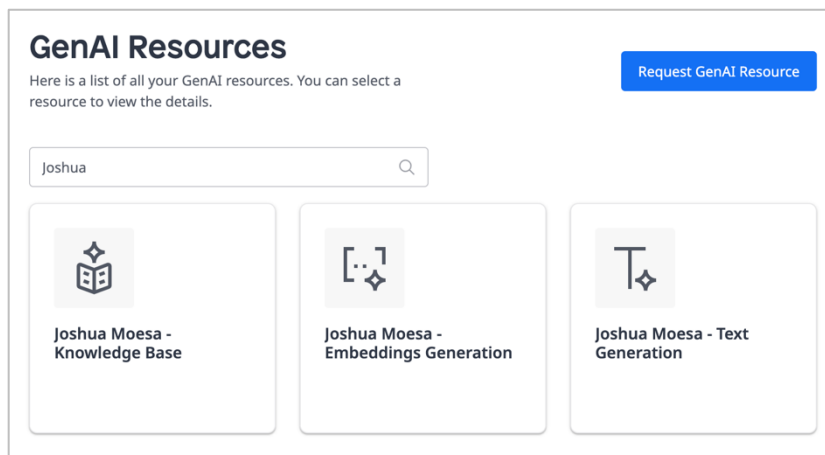
2.1 Mendix Studio Pro

This walkthrough is written for Mendix Studio Pro version 11.12.2.

2.2 Mendix GenAI Resources

In your web browser, go to: <https://genai.home.mendix.com/>. Login with your Mendix account if haven't done that already.

Validate that you have Mendix GenAI resources available in the Mendix GenAI portal. You should see resources *Text generation*, *Knowledge Base* and *Embeddings Generation*

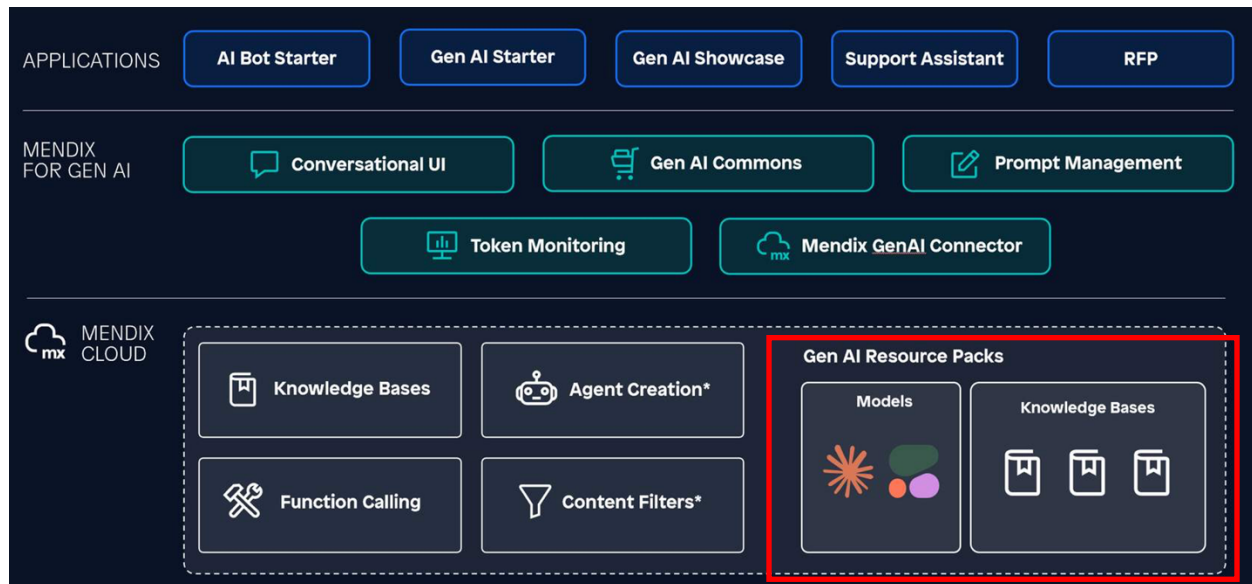


The Mendix GenAI portal is the central hub for configuring and managing Mendix GenAI resources. Target customers are customers open to Mendix Cloud, who do not yet have existing AI infrastructure that they would like to tie into.

Visit the Mendix Docs for more info:

<https://docs.mendix.com/appstore/modules/genai/mx-cloud-genai/resource-packs/>

Great job! You have finished all the prerequisites to gain access to Text Generation, Embeddings and Knowledge base: the Mendix Cloud GenAI Resource Packs. The Text generation resource pack hosts the Large Language Model which we will refer to “model” from now on. The Knowledge base resource pack hosts the Vector database and file store.



3 Exercise – Chatbot extension with RAG

This exercise is designed to introduce how Mendix can be used to build *Smart Apps*. You are going to add AI chatbot functionality to the *Blank GenAI App* project. As a result, users will be able to have a conversational experience.

In summary:

- In the Mendix Cloud Portal, you are going to see how to add content to your Mendix Cloud GenAI Knowledge Base resource.
- In Studio Pro, you will implement the [RAG](#) pattern:
 - o Implement Conversational UI;
 - o Implement a microflow that will wire your Mendix Cloud GenAI configuration to the UI;
 - o Connect your Mendix Cloud GenAI Knowledge Base resource;
 - o Connect your Mendix Cloud GenAI Model resource.



The Mendix Cloud GenAI Knowledge Base is Mendix capability that makes it easier for developers to develop apps using GenAI architectural patterns like RAG and semantic similarity search.

Knowledge bases enable you to bring your own data for RAG, semantic similarity search, and other generative AI use cases:

- Make your app's data available through integration
- Connect to third-party information sources
- Manage knowledge base content and add metadata labels

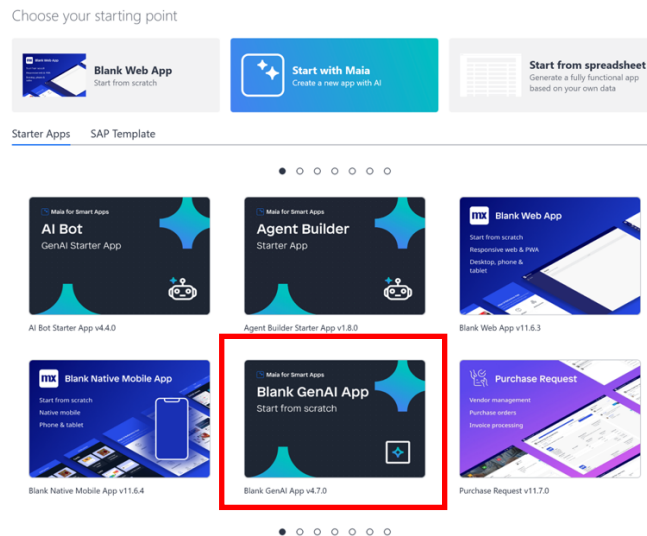
Visit Mendix Docs for more information:

<https://docs.mendix.com/appstore/modules/genai/mx-cloud-genai/resource-packs/>

3.1 Create the app

You will create a new app based on the [Blank GenAI App](#) starter app. This manual is written for version 5.2.0.

1. In Studio Pro, create a new app using the Blank GenAI App starter app.



2. After pulling the code, you will notice a project error in Studio Pro. Go to tab Errors and double-click on the error: *The active configuration contains...Encryption.EncryptionKey.*
3. In *App Settings* pop-up click on the **Default** configuration row.
4. In *Edit Configuration* pop-up click on tab **Constants** and double-click the constant *Encryption.EncryptionKey*
5. In *Edit Constant Value* pop-up in field *Value* enter a 32-character string. For example: `h[<Zrwj;LbGFL_%d1{sQy+>S@T'b3y[C`
6. Check that the error is gone.

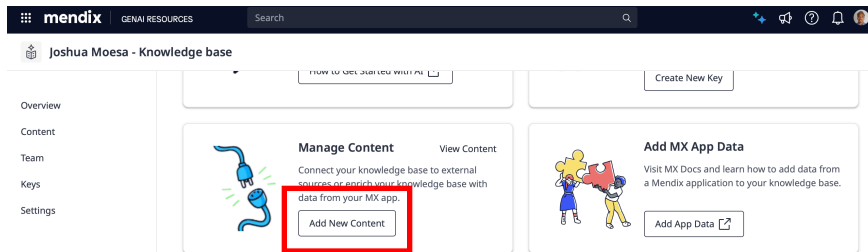
3.2 Add files to the knowledge base



In the workshop this step has been completed by the facilitator. You will skip this step because the knowledge base is shared by all participants. This helps prevent multiple developers from deleting or updating the data at the same time, which could leave the knowledge base in an inconsistent state.

You will add PDF files (bicycle repair manuals) to the Knowledge Base, enabling the chatbot (the bike repair specialist) to incorporate its content into its responses.

1. Go to the Mendix Cloud GenAI Resources page <https://genai.home.mendix.com/> and click on tile <your name> - Knowledge base
2. Click on **Add New Content**.



3. In the pop-up *Bring Your Own Data*, click tile *Add Files Like .TXT or .PDF*
4. In pop-up *Add Files – Add Metadata* enter the following:
 - Key: `type`
 - Value: `manuals`
 - Click *Next*
5. In pop-up *Add Files – Select Collection* select *Default Collection* (“*PortalManagedContent*”) and click **Next**.
6. Upload the PDF files. The status spinners indicate that the Mendix Cloud is processing the files. As a result, each file is:
 - Stored in a file store
 - Chunked in different pieces
 - Each chunk is stored in the vector database as [vector embeddings](#)

Files						
Status	File Name	File Size	Date Added	Collection	Metadata	
✓	E-Bike System Troubleshooting Guide.pdf	982.55 KB	11/20/2025, 11:24 PM	PortalManag...	<code>TYPE: MANUALS</code>	🗑️ 📄
✓	Bike Shop Manual A Guide For Basic Rep...	3.28 MB	11/20/2025, 11:24 PM	PortalManag...	<code>TYPE: MANUALS</code>	🗑️ 📄
✓	Bicycle owners guide to assembly and M...	3.85 MB	11/20/2025, 11:24 PM	PortalManag...	<code>TYPE: MANUALS</code>	🗑️ 📄

3.2.1 Add Conversational UI to the page

You will implement the [Conversational UI](#) module in a new page and implement the microflows that wire your Mendix Cloud GenAI connection configuration to the UI. Enabling users to interact with applications naturally through chat-based interfaces.

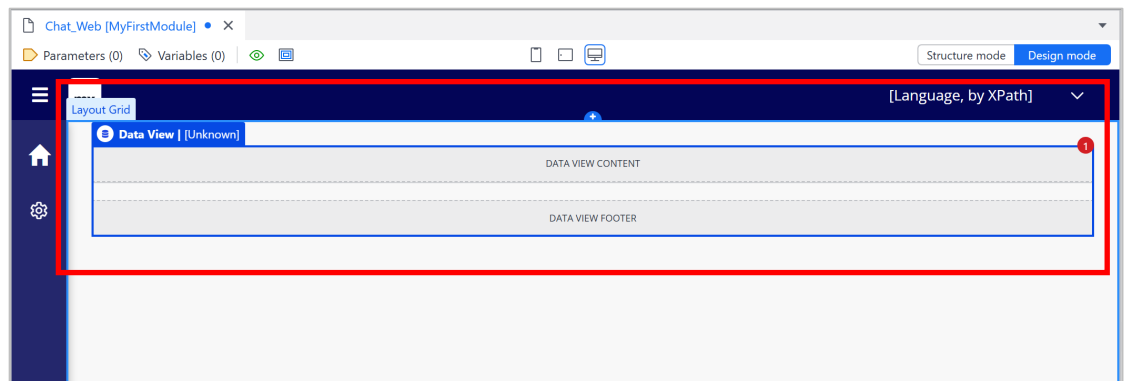


The Mendix Marketplace module *Blank GenAI App* is a starting point to create agentic apps, a conversational UI chatbot or just include GenAI text generation capabilities. The module ships with other marketplace components, such as GenAI Commons, Conversational UI and Mendix Cloud GenAI Connector.

Visit Mendix Marketplace for more information:

<https://marketplace.mendix.com/link/component/227934>

1. Create new blank page *Chat_Web* in module *MyFirstModule*. Add the following:
 - Within the column of the row, add a new Data View.



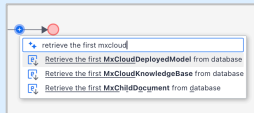
- Configure the new Data View:
 - At Data Source select type *Microflow*.
 - Create a new Microflow:
 - Name: DS_ChatsContext
 - Save it in module *MyFirstModule*.

2. Edit the Microflow you just created: *MyFirstModule.DS_ChatContext*

- Open the microflow and add activity *Retrieve*
 - Source: *From database*
 - Entity: *GenAICommons.DeployedModel*
 - For this exercise, we are going to tie ourselves to a specific LLM using XPath constraint: *Model=*
`'eu.anthropic.claude-sonnet-5'`
 - Options > Range: *First*



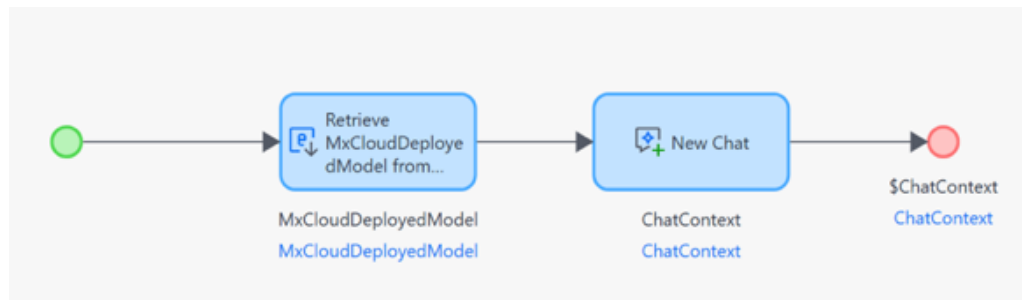
You can add the Retrieve activity even quicker with Maia: click a blue plus-sign and type 'retrieve the first MxCloudDeployedModel from database'. Add the XPath constraint afterwards.



- Add activity *New Chat*
 - Deployed model: `$MxCloudDeployedModel`
 - Action microflow:
 - Click **Select**
 - Create a new microflow:
`ChatContext_ChatWithHistory_ActionMicroflow_MxGen AI_RAG`
 - System prompt:

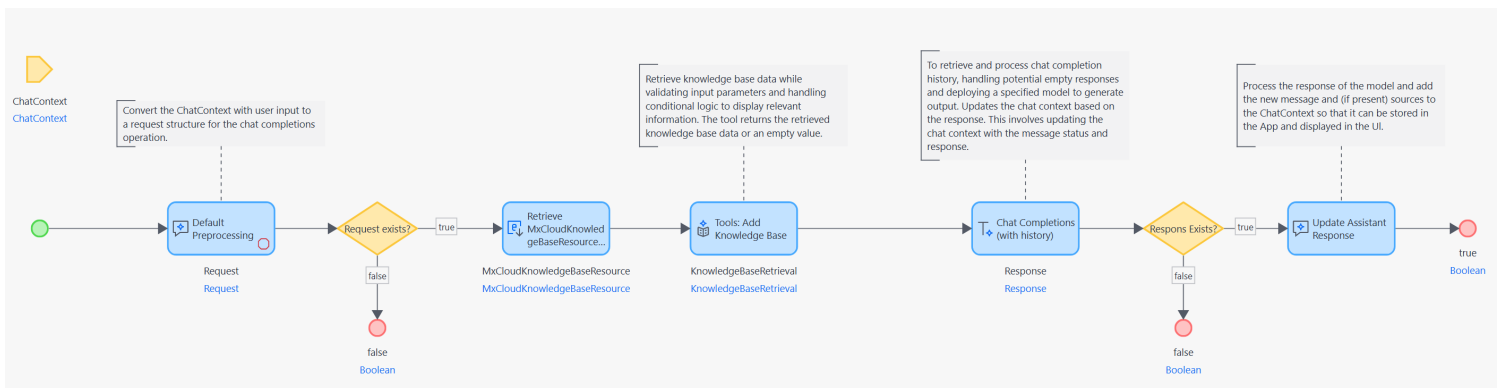
'You are a technical assistant trained to answer questions using a specific set of reference manuals. Use only the provided documents as your source of truth, and prioritize accurate, document-based responses over general knowledge. If the information is not found in the documents, respond clearly that it is unavailable.'
 - Provider name: empty
 - Click **OK**
- Right-click the New Chat activity and click **Set `$ChatContext` as return value**
- Right-click anywhere in the Microflow, then Properties > Security > Allowed roles click button **Select**. Select *User* and *Administrator*.

This is the result:



3. In *New Chat* you referenced a new Action Microflow called: *ChatContext_ChatWithHistory_ActionMicroflow_MxGenAI_RAG*

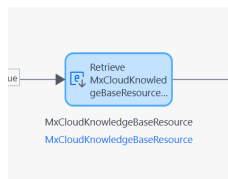
In this Action Microflow you will retrieve the Knowledge Base object that you have access to via de Mendix GenAI Cloud Connector. Then the given user prompt will be augmented by the relevant knowledge base content and then fed to the model. The output of the model will be returned in the chat screen.



There are two ways to finalize this step. Pick one:

Way 1: Use the prebuilt microflow

- Download the SOLUTION ZIP file from the Digital Workshop Room
- Modify the Retrieve MxCloud KnowledgeBaseResource step to your own situation



- Then continue to step 4.

Way 2: Build the microflow from scratch. Continue on the next page.

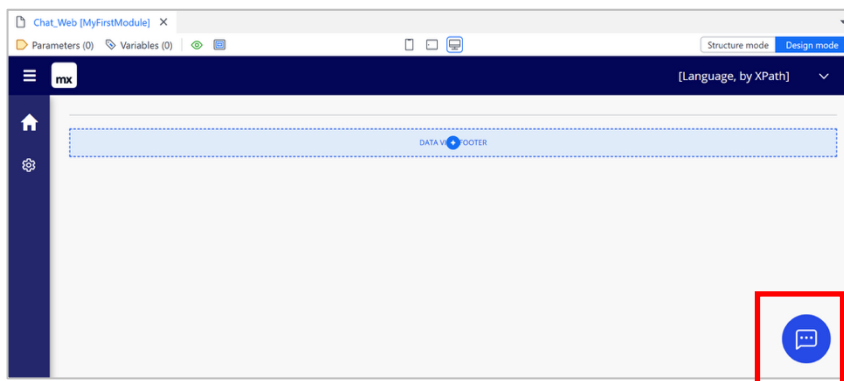
Open the empty Microflow you created earlier and:

- Add a Parameter
 - Data type: *Object*
 - Entity: `ConversationalUI.ChatContext`
- In the flow, add first activity *Default Preprocessing*.
Info: This will convert the ChatContext with user input to a Request structure for the chat completions operation.
- Next, add a *Decision*
 - Caption: `Request exists?`
 - Expression: `$Request != empty`
- Next, add a *Retrieve*
 - Source: From database
 - Entity: *GenAICommons.ConsumedKnowledgeBase*
 - [optional] If you have already access to multiple Mendix Cloud GenAI knowledge base resources, use a XPath constraint for proper filtering, for example:
`DisplayName = 'Knowledge Base'`
 - Options > Range: First
- Next, add a *Tools: Add Knowledge Base*
 - Request: `$Request`
 - MaxNumberOfResults: `5`
 - MinimumSimilarity: `0.2`
 - MetadataCollection: `empty`
 - Name: `'BicycleShop'`
 - Description: `' '`
 - ConsumedKnowledgeBase: `$ConsumedKnowledgeBase`
 - CollectionIdentifier: `'PortalManagedContent'`
 - DisplayTitle: `'Knowledge base search hit'`
 - IsVisible: `true`

- Next, add a *Chat Completions (with history)*
 - Request: `$Request`
 - DeployedModel expression:

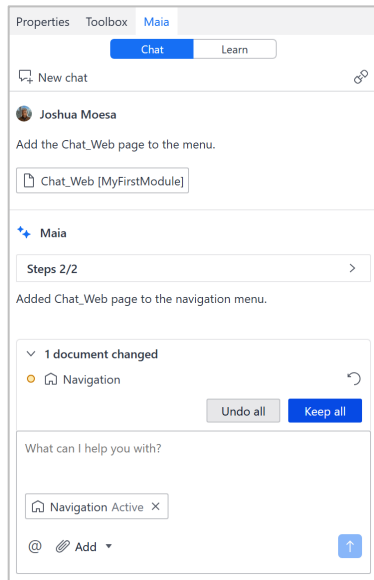

```
$ChatContext/ConversationalUI.ChatContext_ProviderConfig_Active/ConversationalUI.ProviderConfig/ConversationalUI.ProviderConfig_DeployedModel/MxGenAIConnector.MxCloudDeployedModel
```
 - Add activity *Decision*
 - Caption: `Response exists?`
 - Expression: `$Response/GenAICommons.Response_Message != empty`
 - The *false* condition goes to End Event with return value *false*
 - The *true* condition goes to the next activity
 - Add activity *Update Assistant Response*
 - ChatContext: `$Chatcontext`
 - MessageStatus expression:


```
ConversationalUI.ENUM_MessageStatus.Success
```
 - Response: `$Response`
 - Use return variable: *No*
 - Modify all End Events:
 - The main End Event should return Boolean *true*
 - The rest should return Boolean *false*
4. Let's finish up page Chat_Web [MyFirstModule]. In the new Data View you created earlier drag-and-drop *Snippet_FloatingChatButton* from the App Explorer into the Data View content pane.



5. Add the Chat_Web page to the navigation menu. Or ask Maia:

Add the Chat_Web page to the menu Fix the errors (manually).



3.3 Arrange programmatic access from app to Mendix Knowledge Base

Note: If you haven't finished the Knowledge Base exercise in Workbook 1, you need to create a new API key and start with step 3.3.1.

Or else, feel free to reuse the key created in Workbook 1 and continue to step 3.3.2.

3.3.1 Generate the knowledge base API Key

1. Go to the Mendix Cloud GenAI Resources page <https://genai.home.mendix.com/> and click on tile <your name> - **Knowledge Base**
2. Click on **Keys** in the left menu bar. Then click on button **Create Key**
3. In the pop-up, enter a description like your name. Click on button **Generate API Key**.
4. In the next pop-up, copy the generated key and store it in a file on your computer. Click button **Done**.

3.3.2 Use the API key

1. In Studio Pro, click the green play button to run the project locally. Log in with MxAdmin:1.
2. In left-menu bar hover your mouse to the settings wheel and click **Mendix Cloud Configuration**
3. In screen *Mendix Cloud GenAI Configuration* click on tab **Knowledge bases**
4. When the tab *Knowledge bases* is active, click on button **Import key** and paste your API key.
5. Test the connection by clicking **Test Key**. Validate that a success message appears.

3.4 Arrange programmatic access from app to the Mendix Text Generation (model)

Feel free to reuse the key created in Workbook 1 and continue to step 3.5.2.

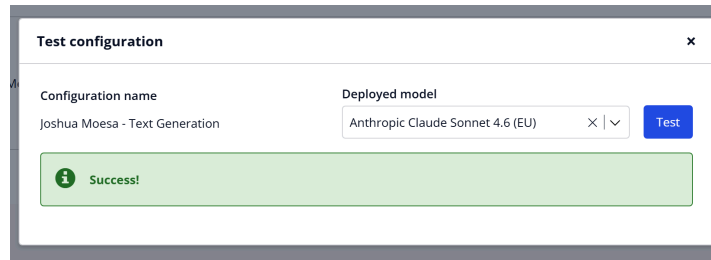
3.4.1 Generate the Text Generation (model) API Key

1. Go to the Mendix Cloud GenAI Resources page <https://genai.home.mendix.com/> and click on tile **<your name> - Text Generation**
2. Click on **Keys** in left menu bar. Then click on button **Create Key**
3. In the pop-up, enter description like your name. Click on button **Generate API Key**.
4. In the next pop-up, copy the generated key and store it in a file on your computer. Click button **Done**.

3.4.2 Use the API key

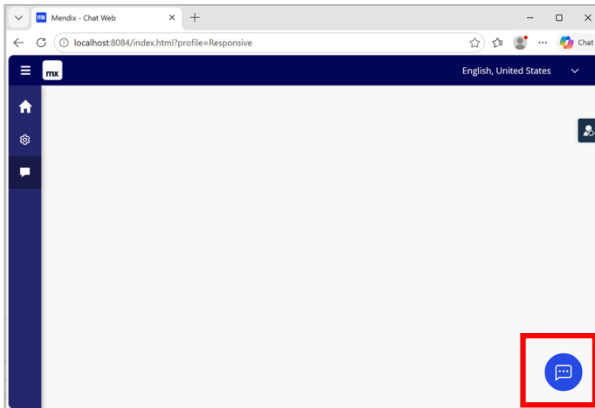
1. If not done already: In Studio Pro, click the green play button to run the project locally. Log in with MxAdmin:1.
2. In left-menu bar hover your mouse to the settings wheel and click **Mendix Cloud Configuration**
3. In screen *GenAI Resources* click on tab **Models**
4. Tab *Models* is active, now click on button **Import key** and paste the key you generated in the previous step.
5. Test the connection
 - Click on the three dots.
 - Then click **Test Key**.
 - Choose a Deployed model. For this exercise, Anthropic Claude Sonnet model is sufficient.

- Click on **Test** and validate success.

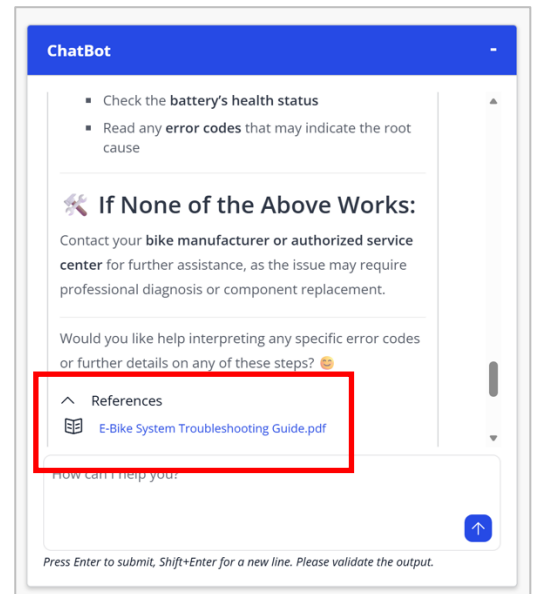


3.5 Test the Chatbot

1. If not done already: in Studio Pro, click the green play button to run the project locally. Log in with MxAdmin:1.
2. Click on the floating chat button on the Chat Web page to start a conversation with the AI chatbot.



3. Validate that based on specific questions, only the relevant response will be returned. And that the references are indicated.



- This question can only be answered using information found in the "E-Bike System Troubleshooting Guide.pdf":

What steps should be taken to troubleshoot an e-bike that does not power on when the battery is fully charged and properly connected?

- Here are two specific questions that can only be answered using information from the "Bike Shop Manual A Guide For Basic Repairs.pdf":

What is the correct method for adjusting cantilever brake pad alignment to ensure they contact the rim properly?

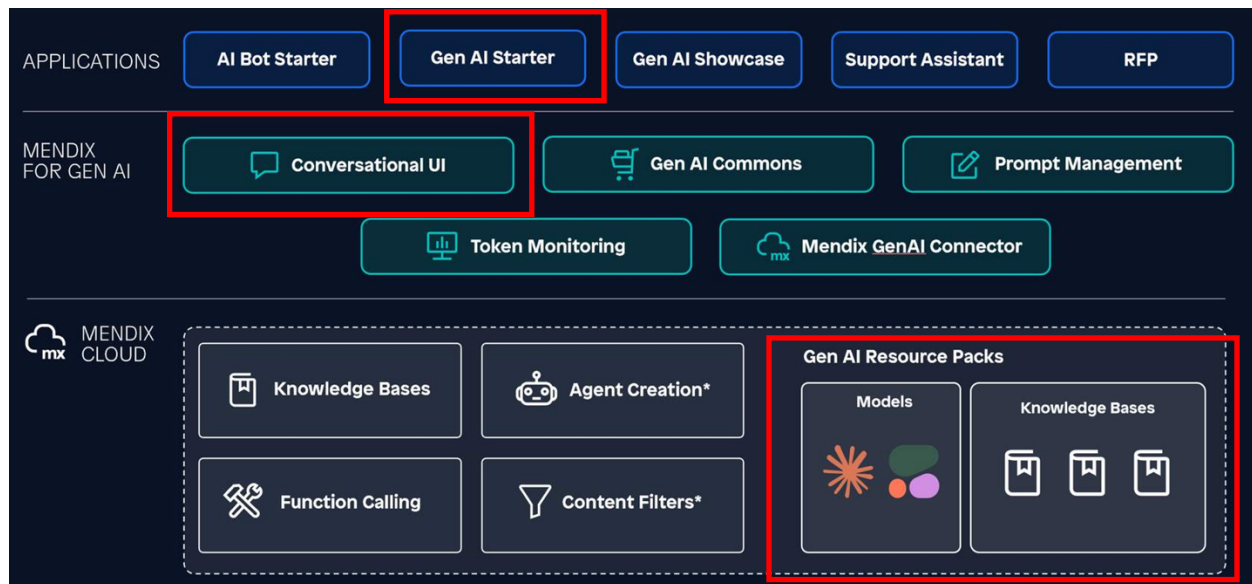
How do you determine and set the correct chain length when installing a new multi-speed chain without referencing the original chain?

- Here are two specific questions that should also be answered using information from the "Bicycle owners guide to assembly and Maintenance.pdf":

Why is it unsafe to inflate bicycle tires using compressed air from a filling station, and what inflation method is recommended?

What are the safety warnings and proper procedures for adjusting saddle height to avoid exceeding the seat post insertion limit?

Excellent! You have finished the exercise. You have wired your AI Chatbot app to Text Generation and Knowledge Base resources. Grounding the chatbots response with your own contextual data.



**Congratulations,
You've Reached
The End Of This
Workbook.**

Change History

Version	Change Date	Log
3.2	2-9-2026	- Upgrade to 11.12.2 - Changes to model selection XPATH constraint
3.1	15-7-2026	- Upgrade to 11.12.1
3.0	8-7-2026	- Upgrade to 11.12.0 - Changed microflows due to multi-model support DS_ChatContext ChatContext_ChatWithHistory_ActionMicroflow_M xGenAI_RAG - Changed validation of importing GRPs due to multi-model support.
2.1	2-6-2026	- Fixed typos - Improved Mendix Cloud GenAI configuration instructions - Added instruction to create Chat_Web page - Added instruction to use Maia for creating a navigation item.